6INTELSY

ENERGY DEMAND FORECASTING + RL LOADING SCHEMING

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Problem

The central challenge in modern energy management is the fundamental misalignment between highly dynamic, household-level electricity demand and the rigid, rule-based scheduling systems currently utilized by grid operators. Because infrastructure planning often relies on historical averages and fixed heuristics, it fails to account for the noisy, non-stationary nature of consumption patterns including seasonal drift and appliance-specific signatures leading to operational inefficiencies and an over-reliance on carbon-intensive peaker plants. This creates a dual-layered technical hurdle: first, the difficulty of generating accurate short-term forecasts from complex temporal data, and second, the optimization dilemma of deferring flexible loads, such as EV chargers or dishwashers, to off-peak hours without compromising user comfort or autonomy.

Datasets

Source of Data

This project utilizes the Individual Household Electric Power Consumption Dataset from the UCI Machine Learning Repository, featuring four years of one-minute interval data from a single residence in Sceaux, France. The raw data, which tracks global power and three specific sub-metering channels, was cleaned and resampled into a working set of 34,168 hourly records. This processed dataset serves as the foundation for forecasting the target variable, Global_active_power, measured in kilowatts.

Features

To capture temporal patterns, the model uses a 12-column feature set that combines seven raw power measurements with five cyclic and binary time-encoding features. By mapping the hour of the day and day of the week to sine-cosine pairs, the system preserves the continuous nature of time, ensuring the model recognizes that hour 23 and hour 0 are adjacent. These features are structured into a 24-step sliding window, where each training sample consists of a 24×12 array of historical data used to predict the Global_active_power for the single subsequent hour.

Test Splits

Splits are strictly temporal to prevent data leakage: no future information appears in training windows, and no training windows overlap with validation or test windows. The dataset is divided as follows: 80% training (27,310 samples), 10% validation (3,393 samples), and 10% test (3,393 samples). A StandardScaler fitted exclusively on the training set is applied to normalize all features before windowing; the same scaler is applied without refitting to validation and test sets.

Data Analysis

Analysis of the household's energy usage revealed a right-skewed distribution with a mean of 1.09 kW, where rare but significant high-demand events occur. Distinct consumption peaks were identified during the morning (07:00–09:00) and evening (18:00–21:00), largely driven by water heating and climate control systems. These specific high-traffic intervals directly dictated the reinforcement learning environment's "must-run" designations to ensure essential loads are prioritized during historically busy hours.

Methods

Long Term - Short Memory

The project uses a sequence-to-sequence LSTM encoder–decoder to model energy consumption patterns. A two-layer LSTM encoder (64 hidden units, with dropout) compresses a 24-hour input into a context vector, which a single-layer decoder uses to predict the next-step forecast. Leveraging LSTM gating mechanisms, the model captures daily and weekly trends and can be extended for multi-step forecasting.

Temporal Convolutional Network

The Temporal Convolutional Network (TCN) provides a parallel alternative to recurrent models using dilated causal convolutions to capture long-term patterns. With four residual blocks and dilation factors (1, 2, 4, 8), it achieves a 31-step receptive field—covering the 24-hour input while preserving causality. Each block uses weight normalization, ReLU, and dropout for stable training, and the final timestep is passed through a linear layer to produce the power forecast.

Natural Language Processor

The NLP component connects system logic with user interaction through two tools: ForecastSummarizer and PeriodClassifier. The summarizer translates forecasting and RL outputs into clear, human-readable explanations while reminding users they can override decisions. Meanwhile, the PeriodClassifier uses TF-IDF and Logistic Regression to categorize user input into peak, off-peak, or low usage. Although trained on a small synthetic dataset, it demonstrates a working proof-of-concept for converting natural language into actionable system inputs.

Reinforcement Learning

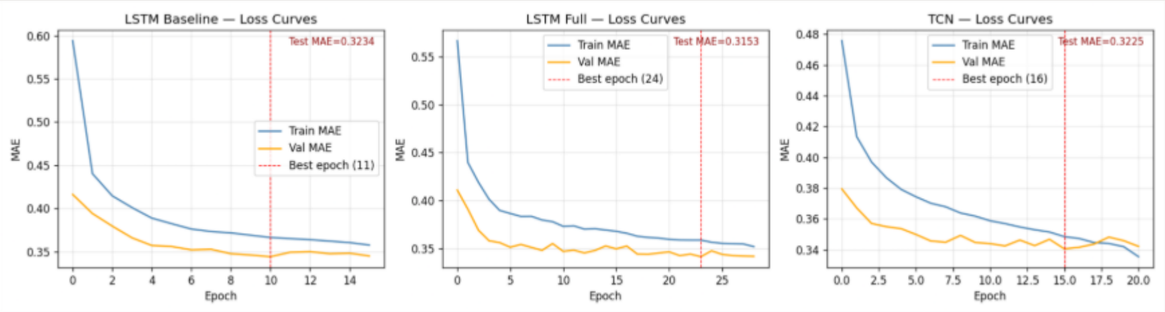
The reinforcement learning component models load scheduling as a Markov Decision Process using a Q-learning agent to balance energy savings and user comfort. In a simulated 24-hour setting, it uses a discretized 4D state (forecast, time, previous load status) to decide whether to run or defer loads. The reward favors cost reduction but penalizes deferring during must-run hours (7, 8, 18–20) and failing to meet a 4-hour minimum runtime. After 500 episodes with ε-greedy exploration, the trained Q-table is saved as JSON for efficient and transparent decision-making.

Results

LSTM Full Encoder-Decoder

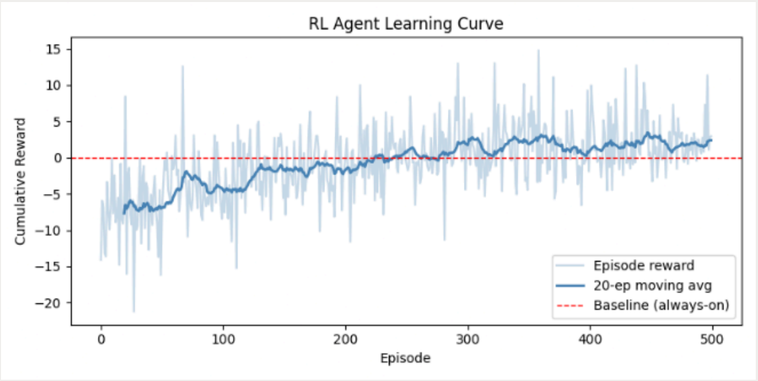
The LSTM Full Encoder-Decoder emerged as the top-performing model with a test MAE of 0.3153 kW, yielding an 18.6% improvement over Linear Regression and a 7.2% gain over Random Forest. While the LSTM Full achieved peak accuracy, the TCN proved highly efficient, reaching a competitive MAE of 0.3225 kW with 33% faster convergence (16 epochs vs. 24). Across the board, deep learning sequence models consistently outperformed tabular baselines, establishing a clear performance hierarchy (Linear < Random Forest < Simple LSTM < Full DL). Ultimately, the narrow gap between the LSTM Full and TCN suggests a performance ceiling for this dataset, with the TCN offering an optimal trade-off for deployments requiring faster training and inference.

Model	Val MAE (kW)	Test MAE (kW)	Best Epoch	Parameters
Liner Regression	0.4022	0.3875	—	—
Random Forest	0.3569	0.3399	—	100 Trees
LSTM Baseline	0.3443	0.3234	11 / 16	10, 401
LSTM Full	0.3414	0.3153	24 / 29	70,465



Reinforcement Learning Results

RL Metrics	Value
Agent AVG Reward (50 Episodes)	2.062 ± 2.885
Always-on baseline reward	0.000 ± 0.000
Reward — First 50 Episodes	-6.415
Reward — Last 50 Episodes	+2.641
Training Episodes	500
Hyperparameters	α=0.10, γ=0.95, ε-decay=0.995



The Q-Learning agent demonstrated a statistically significant learning curve over 50 episodes, transitioning from an initial mean reward of -6.415 to a stable policy of +2.641. This improvement of approximately 9.056 reward units marks a successful shift from penalized random exploration (comfort constraint violations) to a strategy that effectively avoids high-cost deferrals. With a final mean reward of 2.062, the trained agent significantly outperformed the 0.000 "always-on" baseline, confirming its ability to identify optimal off-peak windows for load deferral. While the standard deviation of ±2.885 exists, it reflects the inherent stochasticity of the test-set sequences rather than policy instability, validating the agent's reliability in varied starting conditions.

NLP Results

The PeriodClassifier achieved a perfect 1.00 score in precision, recall, and F1 across its three target categories peak, off-peak, and low-demand within the 22-sample training set. Although the small sample size suggests caution regarding broader generalization, these results confirm that the TF-IDF feature space effectively distinguishes between period types using keyword patterns. Additionally, manual review of the run.py output verified that the ForecastSummarizer consistently generates accurate, grammatically sound, and actionable natural language summaries for all tested scenarios.

Ethics & Policy

EQUITY AND UNEQUAL DISTRIBUTION OF BENEFITS

Automated load scheduling risks marginalizing low-resource households that lack the hardware or technical literacy to shift energy usage, potentially saddling them with higher peak costs.

To address this equity gap, this project integrates a dedicated metric to track the fairness of cost savings across different user profiles and enforces transparency regarding its data's demographic limitations specifically its reliance on a single French household to prevent biased or inappropriate deployment.

USER AUTONOMY

The system prioritizes user control and flexibility, especially for critical and time-sensitive tasks. Certain activities, such as medical equipment use or climate control, require immediate and uninterrupted operation.

Without options like runtime input, deferring, or opt-out, automated decisions could conflict with these essential needs. Therefore, users can override and customize scheduling to ensure safety, comfort, and real-world practicality.

DATA PRIVACY

To mitigate the risk of exposing household routines through high-resolution smart meter data, this project relies exclusively on the anonymized UCI Machine Learning Repository.

By utilizing a .gitignore configuration and a dedicated download script, raw data is kept strictly outside the version control system, ensuring that sensitive information is never committed to the repository or repurposed beyond its intended role in model training and evaluation.

DEMO

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THANKYOU

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